

DHRUV KAURA

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PROFESSIONAL SUMMARY

Software Engineer with experience in Python, FastAPI, and AI-driven backend systems, focused on building scalable applications and intelligent workflow automation solutions. Skilled in REST APIs, asynchronous programming, distributed systems, and LLM-powered orchestration engines. Passionate about Generative AI, prompt engineering, and developing production-ready AI-integrated applications using modern backend and cloud technologies.

EDUCATION

GNA UNIVERSITY

Cumulative GPA: 8.15/10

Phagwara, Punjab

Bachelors of Technology Computer Science & Engineering

2020-2024

WORK EXPERIENCE

PROTOCOLABZ ESERVICES

Phagwara, Punjab

Assistant Web Developer

April 2024-May 2026

- Designed and deployed **RESTful APIs** using **Python** and **FastAPI** serving 3+ production web applications, achieving **~35% reduction in average API response time** via query optimization, connection pooling, and **async I/O** with **asyncio**.
- Built and maintained **backend microservices** integrated with **WordPress** and **Shopify** platforms, resolving **90%+ of API and CMS integration incidents** within the same business day, minimizing client-facing downtime.
- Containerized multi-service infrastructure using **Docker** and **Docker Compose** (API, **PostgreSQL**, **Redis**), delivering consistent parity between development and production environments with **zero environment-related release failures**.
- Optimized **SQL queries** and **PostgreSQL schema design** across multiple services, improving read performance on high-frequency endpoints and ensuring data integrity under concurrent load.
- Automated end-to-end **CI/CD pipelines** using **GitHub Actions**, incorporating lint checks, unit tests, and Alembic database migration workflows, cutting manual deployment effort by **~70%** and eliminating regression releases.

A2IT

Mohali, Punjab

React.js Trainee

June 2023 - August 2023

- Independently designed, developed, and shipped a full-stack Car Service Website using **ReactJS** and **Tailwind CSS** within a 10-week program, applying component lifecycle, Context API, and client-side routing patterns.

CETPA INFOTECH PVT. LTD

Noida, Uttar Pradesh

Web Development Trainee

June 2022 - August 2022

- Built a responsive Coffee Shop Website UI in **ReactJS** using Hooks-based state management, demonstrating component-driven architecture and clean separation of concerns across an 8-week accelerated program.

PROJECTS

REAL-TIME API ORCHESTRATION ENGINE (OrchestrAI) ([Link](#))

- Built an AI-powered workflow orchestration system using **Python** and **LLM-based agents** to automate dynamic multi-step task execution.
- Designed **asynchronous execution** pipelines supporting 200+ concurrent AI/API workflows with fault-tolerant branching logic and real-time task scheduling.
- Implemented modular YAML/JSON-based workflow configurations enabling rapid integration of new AI pipelines and automation flows.
- Optimized orchestration performance using **asyncio** and non-blocking I/O, reducing execution latency by **~40%**.
- Developed scalable backend APIs for workflow management, task monitoring, and intelligent service coordination.

ADAPTIVE WORKFLOW INTELLIGENCE SYSTEM [\(Link\)](#)

- Built a **production-grade asynchronous backend** using **FastAPI** and **PostgreSQL** with **JWT authentication**, **RBAC middleware**, and full audit logging, supporting 5+ distinct workflow types.
- Integrated **Redis** and **Celery** for distributed background task processing with **automatic retry on failure** and real-time task status tracking, enabling decoupled, fault-tolerant job execution.
- Deployed complete infrastructure via **Docker Compose**; implemented **GitHub Actions CI/CD** with automated linting, test execution, and **Alembic** migration workflows, achieving **zero failed deployments** post-launch.

EXPENSE SPLITTER BACKEND SYSTEM [\(Link\)](#)

- Developed a multi-user **RESTful API** in **FastAPI** with group expense management, real-time balance calculation engine, and optimized **SQL queries** handling complex N-party split logic.
- Debugged and resolved **3 critical race conditions** involving transaction atomicity and concurrent balance updates, achieving **100% data consistency** across all test scenarios using database-level locking.
- Implemented **JWT-based authentication** with token refresh, **RBAC**, and endpoint-level authorization middleware, securing all API routes against unauthorized access.

LEADERSHIP & ACTIVITIES

GNA UNIVERSITY COUNCIL OF INNOVATION (GUCI)

Phagwara, Punjab

Coding Head

2021-2022

- Organized **10+ technical events** with **300+ participants** from 5+ universities; managed logistics, problem-setting, and judging criteria for competitive programming rounds.

EDXCELLENCE

Remote

Python and Web Development Mentor

November 2025 - March 2026

- Mentored **30+ students** in Python backend development and web technologies; **90%+** completed hands-on capstone projects, with multiple students securing internships post-program.

TECHNICAL SKILLS

Language: Python(primary), JavaScript (ES6+), SQL

Generative AI & AI Tools: OpenAI APIs, LLM-based Agents, Prompt Engineering, AI Workflow Orchestration, RAG Concepts (Foundational Knowledge)

Backend Frameworks: FastAPI, REST API Design, Asyncio, Celery, SQLAlchemy, Alembic

Databases: PostgreSQL, MySQL, MongoDB, Redis (caching & message broker)

DevOps & Infrastructure: Docker, Docker Compose, GitHub Actions (CI/CD), Git, Postman, Linux

Auth & Security: JWT Authentication, OAuth2, Role-Based Access Control (RBAC)

Concepts: Microservices Architecture, Distributed Systems, Async Programming, System Design, API Rate Limiting, Connection Pooling, Caching Strategies

CS Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, Object-Oriented Programming (OOP)

CMS & Platforms: WordPress, Shopify

CERTIFICATIONS & PUBLICATIONS

Certifications:

- Udemy Full Stack Development by Angela Yu
- AWS Academy Graduate- AWS Academy Cloud Foundation
- UpGrad Python Programming Bootcamp
- UpGrad Generative AI Foundations Certificate

Publications

- Fruits and Vegetable Sanitization and Washing Device(*Dhruv Kaura, Parul Parbhakar, Harjasleen Singh, Dr. Syed Mufassir Yaseen*)